KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

NAME:

TEACHER:

MATHEMATICS

Year 9 5.2

Yearly Exam

2024

Instructions	**Sections and Marks**	
1. Attempt all questions	Part 1 - Financial Maths	/10
2. Show ALL working where needed	Part 2 - SA and Volume	/10
3. A calculator is permitted	Part 3 - Algebraic Techniques and Indices	/10
	Part 4 - Trigonometry	/10
	Part 5 - Data Analysis	/10
	Part 6 - Equations	/10
	Parts 7 - Linear Relationships	/10
	Total marks	**/70**

Part 1: Financial Mathematics

Multiple Choice:

		Mark
1.	What is the weekly wage for Kelly who earns $16.50 per hour and works 35 hours per week? A. $35 B. $577.5 C. $60 D. $1640	1
2.	Calculate Daniels weekly income if he works his usual 40 hour week, at $43 per hour, and decides to work Saturday for 4 hours at time and a half, and Sunday for 3 hours at double time. A. $1820 B. $238 C. $89 D. $2236	1
3.	Tax deductions are: A. The money you receive back if you pay too much tax. B. The money you pay for tax C. A sum that reduces your taxable income and decreases the amount of tax you pay D. The amount that is taken out of your weekly income.	1

Short Answer:

	Question	Working Out	Mark
4	Using the tax table, calculate Sean's taxable income if he earned $165,000 this tax year.		1
5	Using the tax table below, calculate the tax Harry must pay if he earns $70,000, and has $1200 in deductions.		3
6	Using the tax table below, determine if John has a tax debt or refund, and how much is it, based on the following: John earned $36,450 this financial year, his PAYG tax for the year is $5,000.		3

Tax rates for Australian residents

Taxable income	Tax on this income
0 – $18 200	Nil
$18 201 – $45 000	19c for each $1 over $18 200
$45 001 – $120 000	$5092 plus 32.5c for each $1 over $45 000
$120 001 – $180 000	$29 467 plus 37c for each $1 over $120 000
$180 001 and over	$51 667 plus 45c for each $1 over $180 000

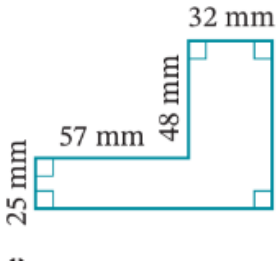
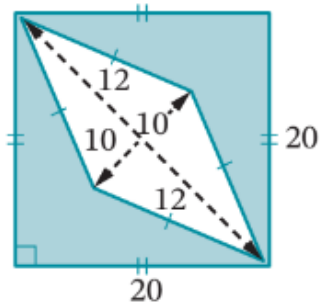
Total /10

Part 2: Perimeter and Area

Multiple Choice:

		Mark
1	How do you convert 1L into mL? A. ÷1000 B. x1000 C. ÷100 D. x100	1
2	A net is what a 3D shape would look like if it were unfolded. A. True B. False	1
3	The volume of a cube is 125 000 cm^3. Convert this to m^3. Select the correct answer. A. 1250 m^3 B. 12.5m^3 C. 1.25m^3 D. 0.125m^3	1

Short Answer:

	Question	Working Out	Mark
4	For the following shape find: a. Perimeter b. Area 	a. Area: b. Perimeter	2 2
	Find the area of the following shape:  Note: Area of a Rhombus formula: $\text{Area} = \frac{1}{2} \times (\text{product of diagonals})$ $A = \frac{1}{2}xy$		3
	Total		/10

	Part 3: Algebraic Techniques and Indices	
	Multiple Choice:	Mark
1	Circle the correct algebraic expression for: *'multiply n by 3 then subtract 5'* A. $5 + 3 + n$ B. $-5 \times 3n$ C. $3n - 5$ D. $n3 + 5$	1
2	If x = 4, c = 7. Calculate y = 8x + 4c using substitution. A. $y = 32$ B. $y = 6$ C. $y = 7$ D. $y = 60$	1
3	Circle which of the following are like terms A. $9y, 8x$ B. $8x^2, 7x$ C. $6g, 7a$ D. $3x, 7x$	1

Short Answer:

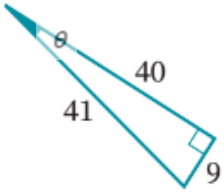
	Question	Working Out	Mark
	Simplify the following expressions:		
4	$2x^2 + 4x + 5x^2$		1
5	$4p \times 3 \times 2q$		1
6	$\frac{24v^2q}{6v}$		1
7	$\frac{3m}{4} - \frac{5m}{12}$		1
8	$\frac{w}{3} \times \frac{3}{4}$		1
9	$\frac{3y}{5} \div \frac{2y}{d}$		1
10	Expand the following: $6(y - 3)$		1
	Total		/10

Part 4: Trigonometry

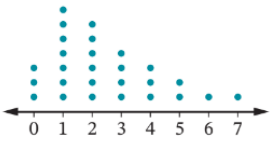
Multiple Choice:

		Mark
1	Circle the 3 names for the sides of a right-angled triangle A. Sin, Cos, Tan B. Opposite, Hypotenuse, Adjacent C. A, B, C D. Sonya, Cassandra, Tanya	1
2	Circle the correct answer for the statement: 'There is 60' in 1°' A. True B. False	1
3	The mnemonic (memory aid) for the trigonometric ratios is which of the following: A. $\frac{O}{H}, \frac{A}{H}, \frac{O}{A}$ B. Opp, Hyp, Adj C. Shine, Call, Tail D. SOH CAH TOA	1

Short Answer:

	Question	Working Out	Mark
4	Label the triangle with the words 'opp, adj, and hyp' with respect to theta: 		3
5	What are the 3 trigonometric ratios	$sin\boldsymbol{\theta}$ = $cos\boldsymbol{\theta}$ = $tan\boldsymbol{\theta}$ =	3
	Evaluate the following expressions to one decimal place:		
6	$sin40°$		1
7	$tan57.4°$		1
8	$4\ cos20°$		1
	Total		/10

Part 5: Data Analysis

	Multiple Choice:	Mark
1	Determine if the shape if positively or negatively skewed a. Positively Skewed b. Negatively Skewed  	1
2	What is the first step to determining the median of a set of data A. Order it in descending order B. Add them all up and divide by 2 C. Order it in ascending order smallest D. Subtract the largest number from the	1
3	Circle which of the following data is 'Categorical' A. Number of girls in class B. Height C. Temperature D. School Reports (A, B, C etc)	1

Short Answer:

	Question	Working Out	Mark
	Determine the following using this set of data: 5 3 8 7 1 6 7 7 5 15 12		
4.	Mean		1
5.	Median		1
6.	Mode		1
7.	Range		1
8.	Using the data, answer the following questions: Stem | Leaf 2 | □ 3 | 0 2 3 4 4 | 1 5 7 8 5 | 2 4 □ 6 9 9 6 | 0 2 3 3 6 8 8 7 7 | 2 5 6 7 7 8 | 2 5	a. What 3 possibilities of numbers could be missing in the '50's'? b. If the range is 63, determine what the missing number is in the '20's' c. Describe the shape:	3
	Total		/10

Part 6: Equations

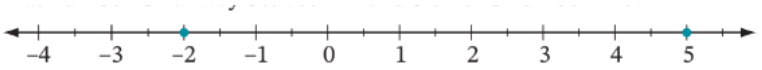
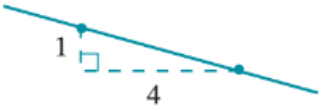
	Multiple Choice:	Mark
1	What can you use to check your work A. Expansion B. Factorising C. Factor Tree D. Substitution	1
2	What is the correct FIRST step to solving the following equation $2(y - 3) = 3(y - 5)$ A. Expand B. Factorise	1
3	Circle the correct equation for the following statement: *When* 7 *is subtracted from* 4 *times a certain number, the answer is* 37. A. $4x - 7 = 37$ B. $7x - 4 = 37$ C. $7 - 4 + x = 37$ D. 37	1

Short Answer:

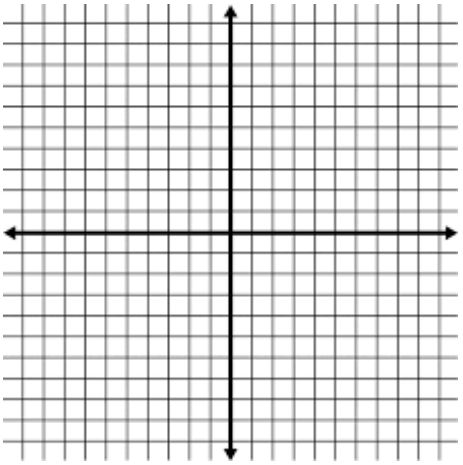
	Question	Working Out	Mark
	Solve the following		
4	$7m + 3 = 10$		1
5	$9 - t = 7t - 2$		1
6	$5(2 + 3p) = -8$		2
7	A catering company charges \$C for a function with P people using the formula C = 1500 + 75.5P		
a	How much does the company charge for a function with 10 guests?		1
b	Find the cost of catering for a group of 60 people.		1
c	Diane has \$10 000 to spend on catering for her wedding reception. What is the maximum number of people she can invite?		1
	Total		/10

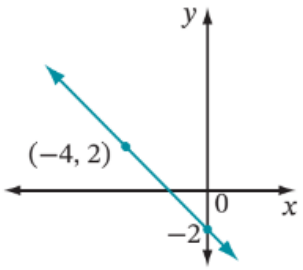
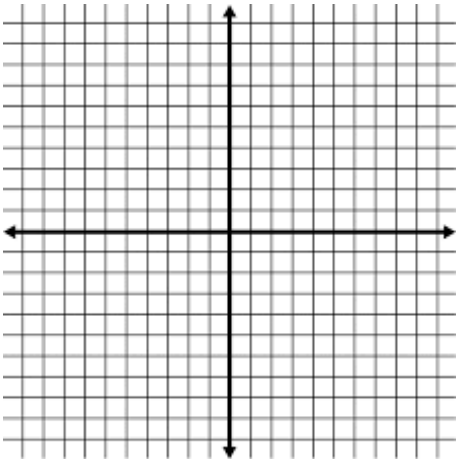
Part 7: Linear Relationships

Multiple Choice:

		Mark
1	What is the midpoint on the number line between -2 and 5  A. 1.5 B. -5 C. 2 D. 0	1
2	The 'run' in, $\frac{Rise}{Run}$ is always: A. Positive B. Negative	1
3	Determine if the gradient is positive or negative A. Positive B. Negative  	1

Short Answer:

	Question	Working Out	Mark
4	Graph the linear equation, y = x – 3, on a number line		2
5	Write the equation of a line if it has a gradient of 2 and a y-intercept of 1.		1

6	Find the equation of the line:  		2
7	Solve the equation 2x – 3 = 5 graphically		2
	Total		/10